

Supplementary Information

When crop rankings do not determine land allocations

Author names and affiliations to be confirmed before submission

Supervisor-review draft

1 Canonical theory and proof record

This supplement summarizes the canonical results used by the manuscript. The complete line-by-line proof package is versioned in `theory/repaiRed/proofs.md`; the theorem transition registry records every replacement of a teacher-Draft claim.

Theorem 1 (Existence and finite-scenario linearity). *If the operational polytope is non-empty and compact, per-acre profits are integrable and the risk-feasible set is non-empty, the loss-CVaR problem has an optimizer. Its feasible set is convex. Under a finite distribution with positive scenario weights, it has the auxiliary linear-programme representation in the main text.*

Proof. Compactness of the risk-feasible subset and continuity of the linear objective give existence. Convexity follows from convexity of loss-CVaR and of the polytope. Substituting the Rockafellar–Uryasev auxiliary function and one non-negative excess variable per scenario gives the finite linear programme [Rockafellar and Uryasev, 2000, 2002]. $\square$

Proposition 1 (Ordinal insufficiency). *A weak or strict suitability ranking alone does not identify a cardinal acreage allocation.*

Proof. A ranking contains only pairwise order relations. Strictly increasing transformations preserve those relations but change cardinal ratios, while arbitrary feasible sets and objectives consistent with the same ranking can have different optimizers. Therefore an allocation requires a mapping, objective, feasible set and selection rule not contained in the order. $\square$

Proposition 2 (Feasibility-forced reversal). *For $s_i > s_j$, if $\sup_{x \in X} x_i + \tau_x < \inf_{x \in X} x_j$, every feasible allocation universally reverses the pair.*

Proof. The premise implies $x_i + \tau_x < x_j$ for every $x \in X$. Every optimizer belongs to X . $\square$

Theorem 2 (Subgradient KKT characterization). *Under a convex constraint qualification, or for a feasible bounded finite-scenario linear programme, a feasible x^* is optimal if and only if there are non-negative multipliers $\lambda, \gamma, \beta, \eta, a, b$ and $d \in \partial r_\alpha(x^*)$ satisfying stationarity and complementary slackness as stated in the main text.*

Proof. Apply convex KKT sufficiency and necessity to the minimization of $-\mu^\top x$. The normal cone of the box contributes $-a + b$; shared restrictions contribute $G^\top \eta$; land and budget contribute $\gamma \mathbf{1}$ and βc . Pairwise subtraction cancels only the common land term. For a finite distribution, a valid CVaR subgradient is constructed from tail weights whose total is one and whose atom weights lie in their permitted intervals. $\square$

Theorem 3 (Restricted dependence ordering). *Fix marginals and a named copula family. If portfolio losses are convex-ordered for every allocation in a declared domain, loss-CVaR is weakly ordered there. If the domain is the whole operational set, the more dispersed model has a subset of the risk-feasible policies and weakly lower optimal expected profit.*

Proof. CVaR is monotone under convex order for integrable losses. Pointwise risk ordering yields nested risk-feasible sets; maximizing the same objective over nested sets yields the value inequality. Crop-specific acreage need not be ordered. $\square$

Proposition 3 (Crossing-set inclusion). *Let g_{ij}^+ and g_{ij}^- be the maximum and minimum pairwise acreage gaps over the optimal set. For any declared selection, the universal reversal region is contained in the selected region, which is contained in the possible region.*

Proof. The selected optimizer is an element of the optimal set. If the minimum gap exceeds tolerance, every element, including the selection, reverses. If the selection reverses, the maximum gap over the set exceeds tolerance. No connectedness or uniqueness of the parameter region follows. $\square$

Theorem 4 (Information actionability). *When ignoring a signal remains feasible, value of information is non-negative. A common posterior-optimal action gives zero value. Informed and uninformed optimized values are weakly non-decreasing under nested action sets, but their difference need not be monotone or strictly positive.*

Proof. The uninformed optimizer is an admissible constant contingent policy, so posterior optimization weakly improves the outer expectation. A common posterior optimizer makes the two objectives equal. Set inclusion weakly raises each maximum separately, but the difference of two weakly increasing functions has no general monotonicity. $\square$

2 Teacher-Draft theorem transitions

The original ranking-reversal equivalence is replaced by the complete KKT identity plus sufficient displacement and feasibility certificates. Scalar tail monotonicity is replaced by the restricted convex-order result. The unique threshold is replaced by crossing sets. Pseudo-diversification is a diagnostic only. Strict information–flexibility complementarity is a simulation hypothesis unless additional lattice and increasing-differences conditions are established.

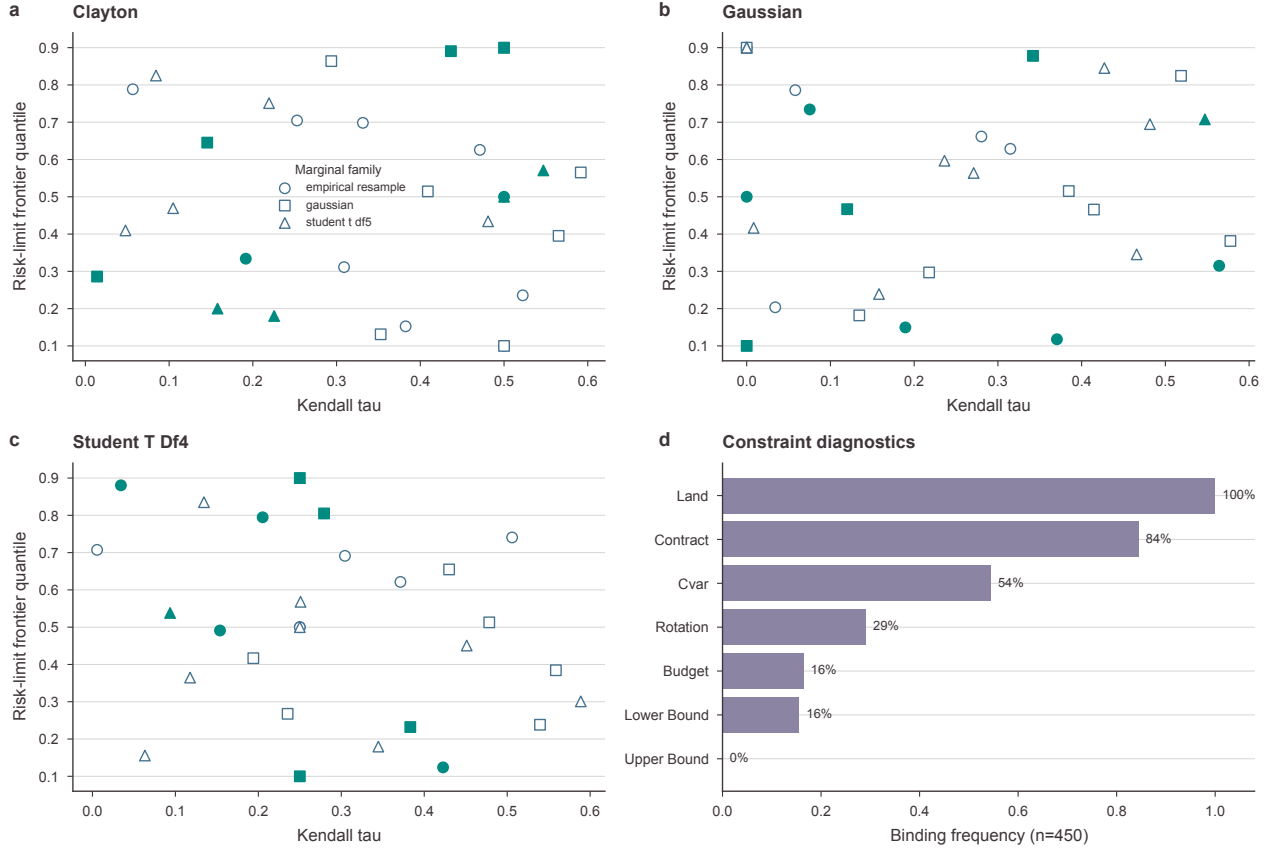
3 Numerical protocol and adverse convergence result

The experiment comprises 90 frozen cells and 450 primary replications. Every scenario matrix is generated from an explicit seed; each optimization records the scenario hash, solver status, objective, allocation, loss-CVaR, optimal-face gaps, named slacks and dual diagnostics. Independent replay reconstructs all tasks in reverse order. Solver sensitivity compares three HiGHS methods at selected anchors.

The optimal-face audit first fixes the primary objective within a declared tolerance and then minimizes and maximizes each relevant acreage gap. This separates possible, universal and selected reversal even when the solver returns a single vertex. Direct loss-CVaR is computed from the optimized loss vector and compared with the auxiliary expression. Tail weights, stationarity, complementarity and feasibility are checked independently.

Figure 1 shows the mixed-factor landscape. Filled markers indicate a universal reversal in at least one of five replications; they are not a fitted decision boundary. Constraint frequencies demonstrate

Mixed-factor simulation landscape: diagnostic only, not a threshold map



Filled symbols indicate universal reversal in at least one of five replications. All simulation patterns are NONHEADLINE because 0/5 convergence rows pass.

Figure 1: Mixed-factor simulation landscape and binding diagnostics. All patterns are nonheadline because the convergence audit fails.

that land and contract restrictions often bind, while the downside-risk constraint is both binding and slack across the design. The pseudo-diversification flag remains descriptive.

The convergence grid uses five scenario counts and ten replications at each count. A row must meet both the frozen numeric-pass fraction and Wilson interval-width requirements. 0 of 5 rows pass. Thus, successful solver checks do not authorize reversal-probability or unique-threshold claims. The complete gate table is available as `tables/supplementary/TableS1_simulation_gates.csv`.

4 Empirical sample, definitions and robustness

The sample flow begins with all parsed state and US-total rows in the frozen official summary. US totals are retained only for the national check. State rows are restricted to complete planted acreage and yield for corn, soybeans and winter wheat in a common state-year. No missing values are inferred. The resulting 231 crop rows form 77 state-years.

Relative yield normalizes each state crop yield by its national same-year value. Standardized revenue multiplies yield by national price. Operating and total-cost margins further subtract corresponding national cost measures. These accounting definitions hold price and cost geography constant, allowing state-yield variation to be examined transparently, but they omit state marketing bases and local costs.

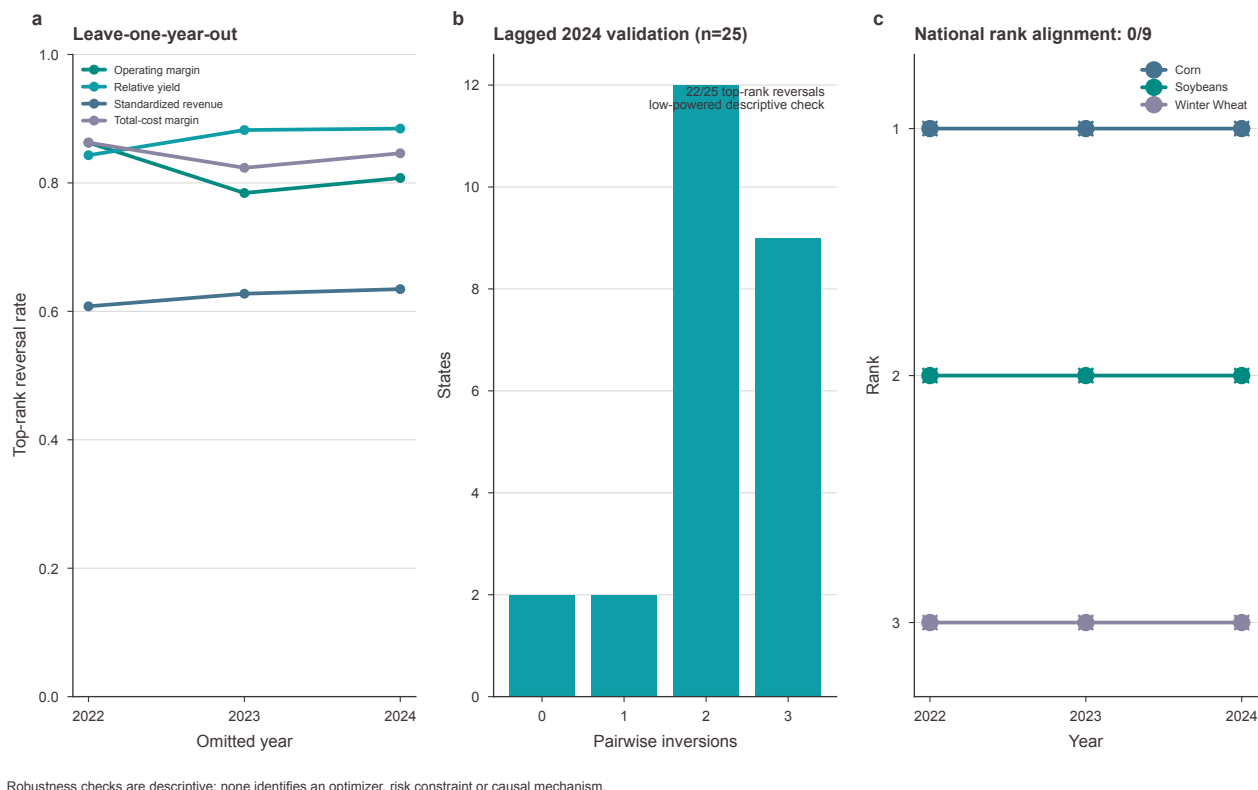


Figure 2: Empirical robustness checks: leave-one-year-out summaries, low-powered lagged validation and national rank alignment.

Figure 2 reports the three main robustness checks. Leave-one-year-out rates remain definition-dependent. The lagged check uses only 2022–2023 relative yields to rank 2024 crops and yields 22 disagreements among 25 states. National margin and acreage ranks align in every comparison. These outcomes are retained together because selective reporting of only the state-level positive pattern would overstate the evidence.

The claim boundary is fixed: state discordance is descriptively identified; accounting consequences are not welfare; geographic patterns are associative; observed-acreage optimality, CVaR binding or causality, copula mechanism, state downside risk and private constraints are not identified. Table S2 in the repository records these statements in machine-readable form.

5 Reproducibility, lineage and artifact inventory

The repository is organized as an evidence graph. Raw data files are immutable and checksum-registered. Processing scripts create canonical panels; formal simulation and empirical scripts write tidy outputs; visualization scripts read only those outputs; manuscript macros read verified summaries. A canonical asset manifest hashes all admitted files and excludes collision-suffixed local copies.

The one-command visual build exports editable PDF/SVG and 300/600-dpi rasters, captions, source-data extracts, grayscale proofs and checksum ledgers. The manuscript build regenerates numeric macros and validates citation keys, section inclusion, claim levels, figure/table paths and closure of the teacher-Draft completion matrix. No teacher-Draft numerical result or legacy threshold is admitted.

Software environment. Python 3.11 is managed with the locked project environment. Optimization uses SciPy/HiGHS; data processing uses pandas and NumPy; figures use Matplotlib with a frozen Nature-style contract. LaTeX source is modular and compiled in a separate isolated build directory.

Artifact availability. The main source is `manuscript/main.tex`; this supplement is `supplementary/supplementary.tex`. Figure and table registries provide generating commands, source data, configurations and checksums. The cover page retains explicit placeholders only for author-owned metadata and declarations that cannot be inferred from scientific evidence.

References

- R. Tyrrell Rockafellar and Stanislav Uryasev. Optimization of conditional value-at-risk. *The Journal of Risk*, 2(3):21–41, 2000. doi: 10.21314/JOR.2000.038. URL <https://doi.org/10.21314/JOR.2000.038>.
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